

System Architecture

High-Level Architecture

```
User
↓
React Frontend
↓
Spring Boot Backend
↓
AI Service Layer
↓
Ollama Local LLM
```

Complete AI Chat Flow

```
User enters prompt
↓
Frontend sends API request
↓
Backend validates request
↓
AI service constructs prompt
↓
Ollama generates response
↓
Streaming response returned
↓
Frontend renders tokens live
```

RAG Flow

```
User uploads document
↓
Backend extracts text
↓
```

```
Text split into chunks
↓
Embeddings generated
↓
Stored in Qdrant
↓
User asks question
↓
Relevant chunks retrieved
↓
Context added to prompt
↓
LLM generates answer
```

Frontend Architecture

```
src/
├─ pages/
├─ components/
├─ features/
├─ hooks/
├─ services/
├─ store/
├─ layouts/
├─ utils/
├─ types/
└─ assets/
```

Backend Architecture

```
src/main/java/com/aiforge/
├─ auth/
├─ chat/
├─ ai/
├─ files/
├─ rag/
├─ memory/
├─ tools/
└─ security/
```

```
|— config/
|— common/
```

Database Architecture

PostgreSQL

Stores:

- users
- chats
- messages
- documents

Qdrant

Stores:

- embeddings
- semantic vectors
- memory vectors

Streaming Architecture

Frontend:

- Fetch streaming
- SSE rendering

Backend:

- Spring WebFlux
- Flux streams
- text/event-stream

Authentication Architecture

```
User Login
```

```
↓
```

JWT Token Generated



Frontend stores token



Token sent in API headers



Backend validates JWT

Deployment Architecture

Frontend → Vercel

Backend → Render/VPS

Database → PostgreSQL

Vector DB → Qdrant

AI Runtime → Ollama